

```

tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ ls -l redirect_*
-rw-r--r-- 1 tarun tarun 126 Oct  7 03:05 redirect_input.txt
-rw-r--r-- 1 tarun tarun 225 Oct  7 03:05 redirect_output.txt
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ cat redirect_output.txt
===== REDIRECTED OUTPUT =====
READ: Hello from redirected input.
READ: This data is being read through STDIN.
READ: This demonstrates how a shell performs input redirection.
=====
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ echo "Practical 10 inode demonstration" > inode_original.txt
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ cat inode_original.txt
Practical 10 inode demonstration
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ ls -li inode_original.txt
74648 -rw-r--r-- 1 tarun tarun 33 Oct  7 03:11 inode_original.txt
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ ln inode_original.txt inode_hard.txt
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ ls -li inode_original.txt inode_hard.txt
74648 -rw-r--r-- 2 tarun tarun 33 Oct  7 03:11 inode_hard.txt
74648 -rw-r--r-- 2 tarun tarun 33 Oct  7 03:11 inode_original.txt
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ ln -s inode_original.txt inode_soft.txt
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ ls -li inode_original.txt inode_hard.txt inode_soft.txt
74648 -rw-r--r-- 2 tarun tarun 33 Oct  7 03:11 inode_hard.txt
74648 -rw-r--r-- 2 tarun tarun 33 Oct  7 03:11 inode_original.txt
74649 lrwxrwxrwx 1 tarun tarun 18 Oct  7 03:13 inode_soft.txt -> inode_original.txt
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ stat inode_original.txt
  File: inode_original.txt
  size: 33          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 74648         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/   tarun)   Gid: ( 1000/   tarun)
Access: 2026-10-07 03:11:49.067364270 +0000
Modify: 2026-10-07 03:11:29.251363639 +0000
Change: 2026-10-07 03:12:33.995362488 +0000
 Birth: 2026-10-07 03:11:29.251363639 +0000
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ stat inode_soft.txt
  File: 'inode_soft.txt' -> 'inode_original.txt'
  size: 18          Blocks: 0          IO Block: 4096   symbolic link
Device: 8,48      Inode: 74649         Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/   tarun)   Gid: ( 1000/   tarun)
Access: 2026-10-07 03:13:08.219362342 +0000
Modify: 2026-10-07 03:13:04.111362312 +0000
Change: 2026-10-07 03:13:04.111362312 +0000
 Birth: 2026-10-07 03:13:04.111362312 +0000
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ find . -inum $(ls -li inode_original.txt | awk '{print $1}')
./inode_hard.txt
./inode_original.txt

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tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ ls -li inode_original.txt inode_hard.txt inode_soft.txt
74648 -rw-r--r-- 2 tarun tarun 33 Oct  7 03:11 inode_hard.txt
74648 -rw-r--r-- 2 tarun tarun 33 Oct  7 03:11 inode_original.txt
74649 lrwxrwxrwx 1 tarun tarun 18 Oct  7 03:13 inode_soft.txt -> inode_original.txt
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ stat inode_original.txt
  File: inode_original.txt
  size: 33          Blocks: 8          IO Block: 4096   regular file
Device: 8,48      Inode: 74648         Links: 2
Access: (0644/-rw-r--r--)  Uid: ( 1000/   tarun)   Gid: ( 1000/   tarun)
Access: 2026-10-07 03:11:49.067364270 +0000
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Change: 2026-10-07 03:12:33.995362488 +0000
 Birth: 2026-10-07 03:11:29.251363639 +0000
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ stat inode_soft.txt
  File: 'inode_soft.txt' -> 'inode_original.txt'
  size: 18          Blocks: 0          IO Block: 4096   symbolic link
Device: 8,48      Inode: 74649         Links: 1
Access: (0777/lrwxrwxrwx)  Uid: ( 1000/   tarun)   Gid: ( 1000/   tarun)
Access: 2026-10-07 03:13:08.219362342 +0000
Modify: 2026-10-07 03:13:04.111362312 +0000
Change: 2026-10-07 03:13:04.111362312 +0000
 Birth: 2026-10-07 03:13:04.111362312 +0000
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ find . -inum $(ls -li inode_original.txt | awk '{print $1}')
./inode_hard.txt
./inode_original.txt
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ echo "Data modified through hard link" >> inode_hard.txt
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ cat inode_original.txt
Practical 10 inode demonstration
Data modified through hard link
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$ cat inode_soft.txt
Practical 10 inode demonstration
Data modified through hard link
tarun@LAPTOP-BRD2JBFC:~/oslab/2520090020_Practical$

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